

Story Agent — Principal Engineer's Elevator Pitch

Audience: hiring manager / principal engineer, job-interview setting **Format:** ~5 minutes spoken (~700 words at a normal 140 wpm) **Goal:** explain the project step by step, the reason for each core feature, and the client benefit.

Deliberated by the OpenRouter crew (Observation Lounge, ~\$0.003) and assembled + fact-checked against the codebase. Verified facts and honest caveats are in the appendix.

The 5-minute spoken script

Read at a normal pace this runs ~4.5–5 minutes. Bracketed cues point at the slide/diagram to show.

[Slide 1 — Control flow]

Every team using an AI coding assistant today pays the same bill: a frontier model — Claude, GPT — charges premium rates for every token, whether it's making an architectural decision or renaming a variable. **Story Agent is my answer to that: a self-hosted autonomous coding assistant where the expensive model does almost none of the work.**

It's built around a metaphor that turned out to be good engineering — an **eleven-member crew of specialized agents**, each with a role, a domain, and *its own* model, that deliberate, plan, and write code together. The premium model is just the orchestrator: it decides and verifies; the crew does the heavy lifting on cheap models.

The problem I'm attacking is twofold. A single frontier model is *expensive*, and it's a *black box* — you can't see why it did what it did, and it doesn't remember what it learned yesterday. I wanted something cheaper, auditable, and that compounds over time.

[Slide 1 — trace the arrow left to right]

Here's the flow. A natural-language request arrives over the **Model Context Protocol** — the open standard — so the same crew plugs into Claude Code, Cursor, or Copilot unchanged. The orchestrator hands it to the **crew mission pipeline**: one agent distills the goal, another assembles the right specialists, and a cost officer assigns each specialist **the cheapest model that can actually do their part**. Most work lands on DeepSeek or Llama; the frontier Anthropic model is reserved for the hardest architecture and security calls only. The specialists deliberate in the open — every position and tradeoff recorded — and produce a *costed* mission plan. For the actual coding, an agentic loop reads, edits, and runs commands on that same cheap model.

[Slide 2 — governance + memory wrap]

Two things wrap every action. First, a **security gate**. It brokers every credential — secrets are never logged — and it classifies every file-and-shell operation green, yellow, or red. A risky command isn't just blocked; it's **auto-remediated** — paths clamped into the workspace, a `--force` flag downgraded — so the agent keeps moving *safely* instead of stalling. Second, **memory**: before acting, the crew recalls what it learned on past missions from a vector database; afterward, it stores its conclusions. So the system gets better the more you use it.

[Slide 3 — the money slide: cost split + routing]

The reason for all of this is one lever: **cost, with proof**. Because the premium model only orchestrates — and I *measure* exactly when the cheap crew is driving versus the expensive one — a full eleven-member deliberation costs a fraction of a cent, on the order of two-tenths of a cent, instead of a frontier-model bill. And every one of those decisions lands on an audit trail.

For a client, that's three concrete things. **One — cost control you can see**: savings versus a frontier baseline are computed and reported, not promised. **Two — governance**: every credential access and every autonomous action is audited and policy-gated, which matters for regulated work, and clients are isolated with their own policies and their own memory. **Three — it compounds**: knowledge persists across sessions and even across *different* AI tools, because the memory is the portable brain, not the vendor.

[Slide 1 — return to the whole]

I'll be straight about maturity. It runs today — the crew deliberates and writes code — and I've defined a **shadow test** to decide when it's reliable enough to fully replace the premium driver; that bar isn't cleared yet. But the architecture is the point: **keep the expensive model thin, delegate to a cheap, governed, self-improving crew, and make the cost of every decision visible**. That's a pattern any team drowning in AI spend can adopt — and it's the same discipline I bring to systems generally: push work to the cheapest place that can do it correctly, and make correctness and cost observable.

(≈ 690 words)

Supporting diagrams

Diagram 1 — Control flow (the whole system in one picture)

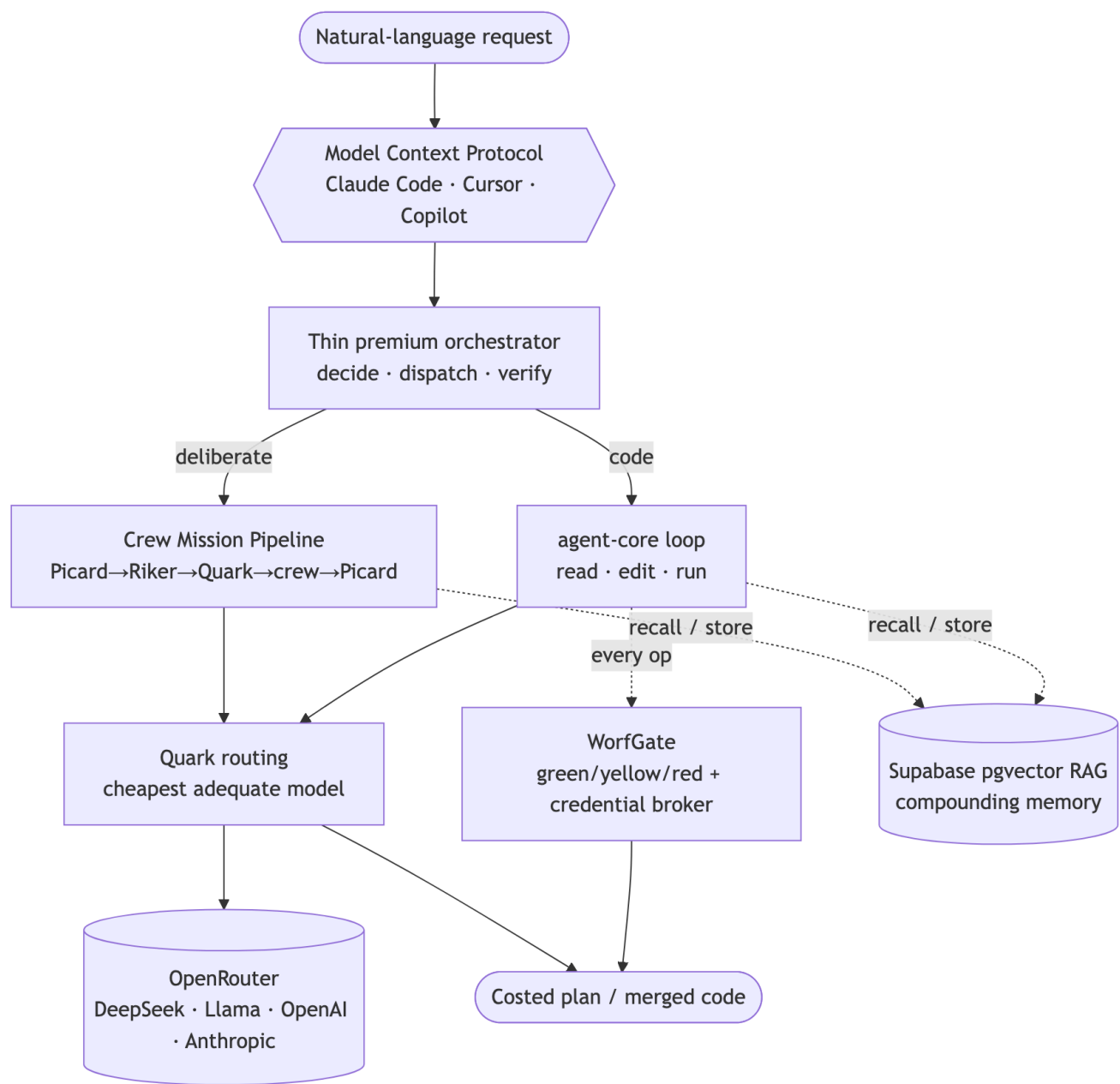


Diagram 2 — Two-lane cost model (why it's cheap)

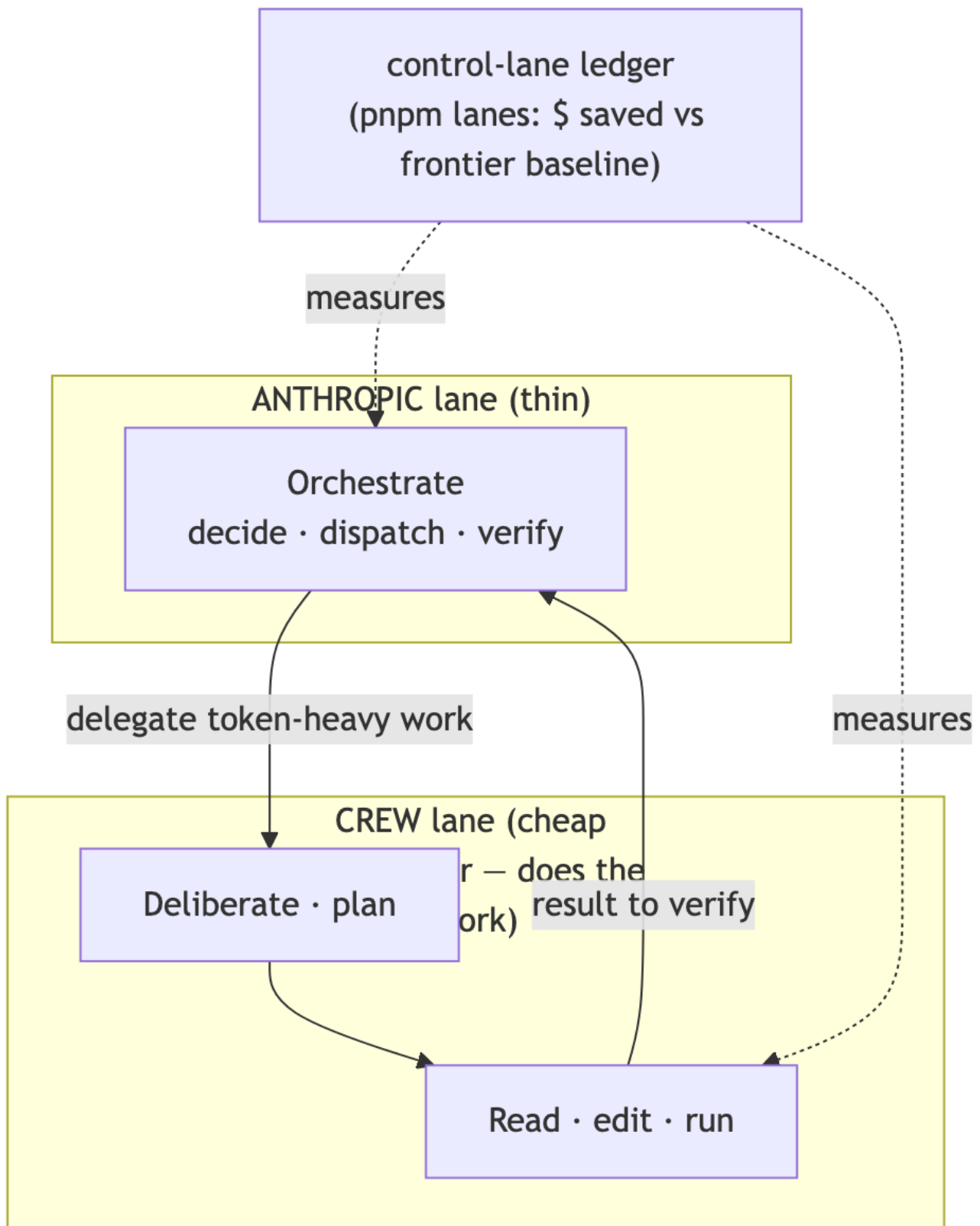


Diagram 3 — Quark's routing decision (cheapest adequate model)

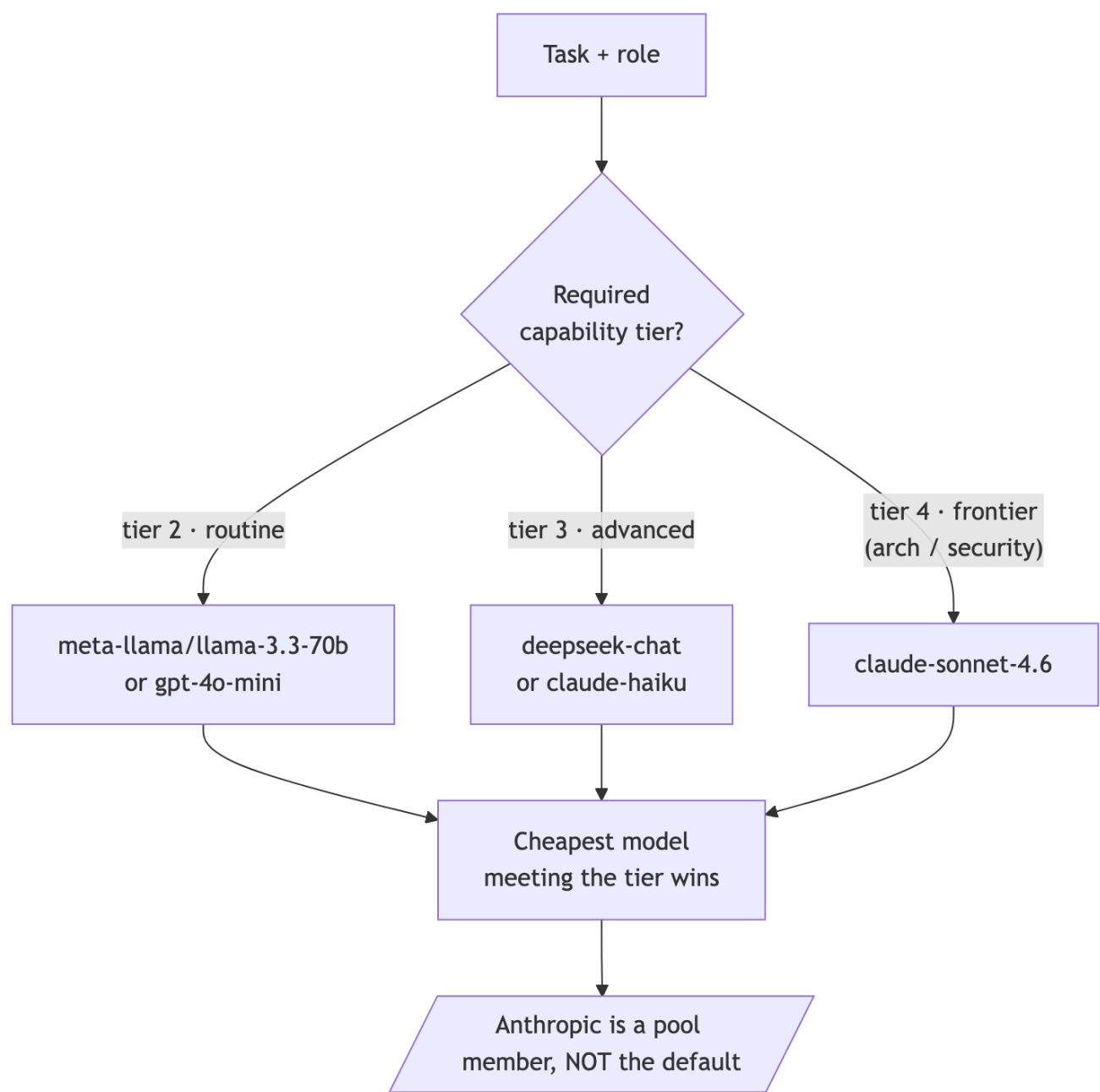
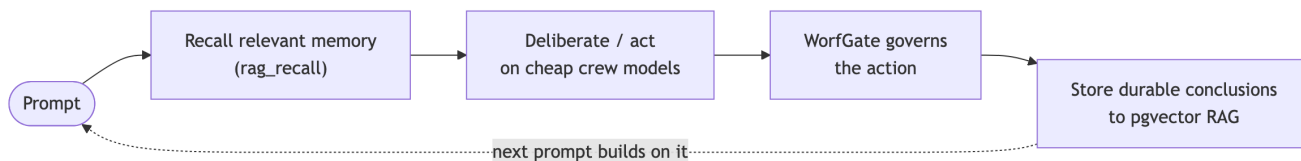


Diagram 4 — The compounding loop (why it gets better)



Why each core feature exists (the "reason" behind the pitch)

Feature	What it is	Why it exists (the reason)	Client benefit
Crew mission pipeline	6-stage deliberation → owned, costed plan	Structured multi-perspective reasoning beats one generalist; plan is auditable	Better decisions, on the record
agent-core loop	Agentic read/edit/run on a cheap model	The heavy coding work must not run on the premium lane	Lower cost per change
Quark routing	Cheapest adequate model per task; Anthropic only tier-4	Cost is the core lever; frontier only where it's warranted	Pay frontier prices only when it matters
WorfGate	Credential broker + green/yellow/red op governor	Autonomy needs accountability; remediate instead of hard-block	Safe automation; audit trail for regulated work
RAG memory	pgvector recall→act→store loop	A system compounds only if each turn builds on the last	Gets smarter with use; portable across tools
Control-lane	Observable crew-vs-Anthropic cost attribution	If cost is the lever, it must be measurable	Savings you can see, not just promised
Client registry	firm→client→project hierarchy, per-client policy + isolation	Multi-tenant, governed by construction	Client isolation + policy floors
MCP server	Exposes the crew's tools to any AI client	The crew brain should be portable, not locked in	No vendor lock-in

Interview framing (what the manager is really testing)

- **Judgment / tradeoffs:** the honest answer to "*why not just use Claude Code?*" — you still use the frontier model, but only as a thin orchestrator; the win is delegating token-heavy work to a cheap, governed, self-improving crew and **measuring** the split.
- **The one memorable line:** "*Keep the expensive model thin; make the cost of every decision visible.*"
- **The one honest line (earns trust):** "*It runs today; the shadow test that says it's ready to fully replace the premium driver isn't cleared yet — and I can tell you exactly what that bar is.*"

Appendix — verified facts & honest caveats

Verified (safe to state): 11-member crew; monorepo of 4 packages (`mcp-server` , `shared` , `ui/Next.js`, `vscode-extension`); ~61k lines of TypeScript; ~150 MCP tool registrations; 41 skill-theory definitions; Model Context Protocol SDK `^1.12.0` ; multi-provider `MODEL_P00L` (Meta,

OpenAI, DeepSeek, Anthropic, Google); pgvector RAG (64-dim, IVFFlat cosine); WorfGate credential broker + green/yellow/red governor; control-lane cost ledger; deployed on AWS Fargate (terraform + docker). Crew deliberations measured at ****~0.002–0.0028 ****; *anInnovationLounge*run **** 0.04****; embeddings ****~\$0.02 / 1M tokens****.

Do NOT overclaim (per code recon):

- The **shadow test** thresholds ($\geq 90\%$ auto-recovery, $\leq \sim 80\%$ of frontier cost) are a *documented go-criterion*, not an achieved result — present as methodology.
- Cost figures are **per-deliberation micro-costs**, not audited monthly client savings; don't generalize "\$0.04" beyond an Innovation Lounge run.
- **No verified production customers** — "Jonah"/"Bayer" are illustrative clients in docs.
- The **README still describes the older Aha story-delivery product**; the crew/OpenRouter identity lives in CLAUDE.md/AGENTS.md — be ready to explain the evolution if asked.
- Minor canon note: the crew includes **Quark (Deep Space Nine)**, not strictly TNG.